

Lakshmi Naga Hrishitaa Dharmavarapu

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EDUCATION

University of Illinois at Chicago
Master of Science in *Computer Science*

Aug 2024 – May 2026
Chicago, IL

KL University
Bachelor of Technology in *Computer Science*

Jul 2019 – Mar 2023
Hyderabad, India

TECHNICAL SKILLS

ML & AI: Python, PyTorch, TensorFlow, scikit-learn, XGBoost, Keras, NLP, spaCy, LLM Applications, RAG, LangGraph, LangChain
Data Science: SQL, Feature Engineering, Statistical Modeling, Hypothesis Testing, A/B Testing, Causal Inference, pandas, SciPy
MLOps & Infrastructure: MLflow, Docker, Kubernetes, FastAPI, CI/CD, REST APIs, Git, Linux, Terraform, AWS CDK
Data Engineering: ETL Pipelines, Spark, Hadoop, Elasticsearch, MySQL, AWS (S3, EC2, RDS, Redshift)
Visualization: Tableau, Looker Studio, Streamlit, Matplotlib

PROFESSIONAL EXPERIENCE

Research Assistant
University of Illinois at Chicago

May 2025 – Present
Chicago, IL

- **IDIATER (Infrastructure and Decision Intelligence for Accelerating Technology & Energy Resilience) Research Group**
- **ML Data Pipeline:** Engineered an end-to-end ingestion pipeline to collect, merge, and structure heterogeneous data — sustainability plans, emissions inventories, and census data — across **100+ municipalities**.
- **NLP & LLM Extraction:** Built a prompt-engineered LLM document parsing system converting raw municipal PDFs into structured datasets with **95%+ extraction accuracy** using NLP extraction chains.
- **Experiment Tracking & Feature Engineering:** Tracked model runs with **MLflow** and transformed **100+ municipal financial reports** into analysis-ready feature sets; built **Streamlit** dashboards presented to urban planning groups, the **Metropolitan Mayors Caucus**, and at the **SRI Congress** spanning **80+ countries**.
- **Research Impact:** Research accepted to the **Sustainability Research and Innovation Congress (SRI)** spanning **80+ countries**; findings adopted by local governments to benchmark climate investments.

Associate Software Engineer
OpenText Technologies

Oct 2022 – Jun 2024
Hyderabad, India

- **Client-Facing POC to Production:** Presented analytics platform demos to **enterprise clients**, gathered feedback, and iteratively evolved POCs into a production-grade **Python + FastAPI** platform on **AWS** serving **100+ stakeholders**.
- **Full-Stack Analytics Platform:** Built **React** frontend over a **FastAPI** backend delivering real-time analytics dashboards to enterprise clients, owning every layer from UI components to API design and deployment.
- **ML-Backed Analytics Platform:** Architected a scalable service layer replacing ad hoc SQL/Excel workflows; incorporated client feedback across iteration cycles to align product behavior with enterprise business needs.
- **Container Orchestration:** Deployed and managed containerized services using **Kubernetes** and **Docker**, ensuring high availability and scalable infrastructure across development and production environments.
- **Large-Scale ETL:** Built **Spark** and **Hadoop** pipelines processing up to **1 TB** via **AWS S3**, automating PDF report generation and eliminating **20+ hours/month** of manual effort.
- **Search & Inference Optimization:** Integrated **Elasticsearch** to offload high-latency queries, reducing response latency by **35%** for a system serving **2M+ users**.
- **Pipeline Validation Automation:** Automated end-to-end data validation and UI testing workflows, cutting manual testing by **60%** and reducing production release risk by **50–60%**.

Software Engineering Intern
OpenText Technologies

Aug 2022 – Sep 2022
Hyderabad, India

- **Data Integrity & Query Optimization:** Maintained and optimized **MySQL** schemas using joins, indexes, and migrations to ensure data integrity; resolved **~40%** of cross-layer issues across search and database components.
- **Deployment & CI/CD Support:** Assisted in deploying containerized services to **AWS** using **Docker** and contributed to **CI/CD** pipeline setup, supporting reliable and repeatable production deployments.

PROJECTS

CommuteGenie — Multi-Agent AI Transportation Assistant — [GitHub](#)

Spring 2026

- **Multi-Agent System Design:** Designed a manager-worker architecture with specialized **Bus, Train/Metro, Context, and Critic agents** orchestrated via **LangGraph/LangChain** and a **ReAct-style planner** for modular, testable agent components.
- **RAG & Retrieval Pipeline:** Implemented a **Chroma/FAISS** RAG pipeline with real-time transit, weather, and disruption signal ingestion; Critic-agent validation reduced hallucinated routes and grounded outputs in retrieved context.

- **End-to-End Delivery:** Deployed a **React + Next.js + Tailwind CSS** frontend over a **Python FastAPI** backend, supporting natural-language commute queries with structured, grounded responses.

NIJ Recidivism Forecasting — ML + Algorithmic Fairness Pipeline — [GitHub](#)

Fall 2025

- **Survival Model Training:** Trained 3 conditional **XGBoost** survival models (Year 1/2/3) on **25,835 individuals** using RandomizedSearchCV (40 iterations, 5-fold StratifiedKFold) with scale_pos_weight for class imbalance.
- **Responsible AI & Fairness Audit:** Measured FPR/TPR/calibration gaps across Race, Gender, and Race×Gender subgroups with 500-sample bootstrap CIs; found gender gap (10.5-pp) exceeded racial gap (2.8-pp) — counter to COMPAS-style findings.
- **Bias Mitigation:** Benchmarked Kamiran-Calders reweighing, ThresholdOptimizer, and ExponentiatedGradient; detected proxy encoding via **SHAP** correlation analysis to ensure equitable model deployment.